An obstruction to the proposed binary Hankel product formula

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Abstract

The fixed-offset product formula in Sobolewski–Ulas's Conjecture 4.6 fails already at offset 2. We evaluate that entire subsequence using their determinant recurrence, then rule out every allowed shift and prefactor. The obstruction persists even when polynomial prefactors are replaced by arbitrary rational functions.

Task 1. Full disproof

Let $s(a)$ be the number of ones in the binary expansion of a, and put

$$\mathcal{H}(n) = \det\big[s(i+j)\big]_{0\le i,j<n}, \qquad B_m = \prod_{j=1}^{m} (2^j-1)^{2^{m-j}}, \qquad T_m = \sum_{j=1}^{m} \frac{2^{j-1}}{2^j-1}.$$

Thus $\mathcal{H}(n) = \mathcal{H}(n,1)$ in the source's notation. The conjecture asserts that, for each fixed $k \ge 0$, there exist a fixed integer $r \ge 0$, a sign ε, a polynomial $P \in \mathbb{Q}[X]$, and a constant $c \in \mathbb{Q}$ such that eventually

$$\mathcal{H}(2^m + k) = \varepsilon P(2^m)(c + T_{m-r})B_{m-r}. \tag{1}$$

The equality is required for all sufficiently large integers m.

Theorem 1. *For every $m \ge 2$, with $X = 2^m$,*

$$\boxed{\mathcal{H}(2^m+2) = \frac{B_m(2+T_{m-1})}{(2^{m-1}-1)(2^m-1)} = \frac{B_m}{A(X)}\big(T_m + d(X)\big),} \tag{2}$$

where

$$A(X) = (X/2-1)(X-1), \qquad d(X) = \frac{3}{2} - \frac{1}{2(X-1)}.$$

There are no $r \ge 0$, $c \in \mathbb{R}$, and nonzero $P \in \mathbb{R}(X)$ for which (1) holds eventually at $k = 2$. Consequently Conjecture 4.6 is false.

Exact evaluation

We use the proved recurrence in [1, Theorem 4.2 and Proposition 4.3]. Here are its specializations, with all normalizations specified. For $2^a < n \le 2^{a+1}$ let $h(n,x)$ be the Hankel determinant of the weighted digit sum with weights $(1,\dots,1,x)$, where x is the weight at position $a+1$. Let $g(n,x)$ be

the determinant of the $n \times n$ matrix whose first $n-1$ columns have entries $s_{\mathbf{w}}(i+j+1) - s_{\mathbf{w}}(i+j)$ and whose last column is 1. For $m \geq 2$ set

$$U_m(x) = h(2^m - 1, x), \qquad V_m(x) = g(2^m - 1, x).$$

Direct 3×3 determinants give $U_2(x) = 3 - x$ and $V_2(x) = 2 - x$. The source recurrence, applied on its second branch, gives for $m \geq 3$

$$\begin{aligned} V_m(x) &= C_m(x) V_{m-1}(x/2), \\ U_m(x) &= C_m(x) \left(U_{m-1}(x/2) + \frac{V_{m-1}(x/2)}{2 - x} \right), \\ C_m(x) &= 2^{\,2^{m-1}-3} (2 - x)^{2^{m-1}}. \end{aligned} \tag{3}$$

These identities follow with $n = 2^m - 1$ and $a = m - 1$; in particular $(-1)^n = -1$. Dividing the first two identities and iterating, initially as identities of rational functions, gives

$$\frac{U_m(x)}{V_m(x)} = 1 + \sum_{j=0}^{m-2} \frac{1}{2 - x/2^j}. \tag{4}$$

Iteration of the first identity in (3) at $x = 1$ also gives

$$\begin{aligned} V_m(1) &= \left(2 - 2^{-(m-2)}\right) \prod_{j=0}^{m-3} 2^{\,2^{m-j-1}-3} \left(2 - 2^{-j}\right)^{2^{m-j-1}} \\ &= (2^{m-1} - 1) \prod_{j=1}^{m-2} (2^j - 1)^{2^{m-j}} = \frac{B_m}{(2^{m-1} - 1)(2^m - 1)}. \end{aligned} \tag{5}$$

The product is empty at $m = 2$. To check the powers of 2 in the second equality, their total exponent is

$$-(m-2) + \sum_{j=0}^{m-3} \big((1-j) 2^{m-j-1} - 3 \big) = 0.$$

This is the elementary finite geometric-sum identity $\sum_{j=0}^{m-3} (1-j) 2^{m-j-1} = 4m - 8$. Equation (4) at $x = 1$ is $U_m(1) = V_m(1)(1 + T_{m-1})$.

Finally use the source recurrence's first branch with $n = 2^m + 2$, $a = m$, and all weights 1. It is applicable for every $m \geq 2$. Its scalar prefactor and upper-right matrix entry are both -1, so

$$\mathcal{H}(2^m + 2) = U_m(1) + V_m(1).$$

Together with (4)–(5), this proves the first equality in (2). The second follows from $T_m - T_{m-1} = X/(2(X-1))$. For example, the values at $m = 2, 3, 4, 5$ are $3, 11, 2403, 1127461923$.

All fixed shifts and prefactors are impossible

The positive constant

$$\lambda = \sum_{j=1}^{\infty} 2^{-j} \log(2^j - 1)$$

is finite, since the summands are bounded by $j2^{-j}\log 2$, and it is positive already from $j = 2$. Hence

$$\frac{\log B_m}{2^m} \longrightarrow \lambda.$$

Also

$$T_m = \frac{m}{2} + \frac{1}{2}\sum_{j=1}^{m} \frac{1}{2^j - 1} = \frac{m}{2} + O(1). \tag{6}$$

By (2), $2^{-m}\log \mathcal{H}(2^m+2) \to \lambda$. If (1) holds with any nonzero rational function P, then $\log|P(2^m)| = O(m)$ and $\log|c + T_{m-r}| = O(\log m)$. Thus the same limit computed from (1) is $2^{-r}\lambda$. It follows that $r = 0$.

Absorb the sign into P and define $R(X) = A(X)P(X)$. Using (2), the remaining equality becomes

$$(R(2^m) - 1)T_m = d(2^m) - cR(2^m). \tag{7}$$

If $R \not\equiv 1$, its difference from 1 has only finitely many zeros and poles. Consequently (7) would express T_m eventually as $F(2^m)$ for the fixed rational function

$$F(X) = \frac{d(X) - cR(X)}{R(X) - 1}.$$

This is impossible by (6): a rational function at infinity either has a finite limit, or is asymptotic to bX^e for some $b \neq 0$ and positive integer e. Neither behavior at $X = 2^m$ is asymptotic to $m/2$. If $R \equiv 1$, (7) instead forces $c = d(2^m)$ for every sufficiently large m, also impossible because $d(2^m)$ is strictly increasing. This proves the theorem.

The argument concerns the exact eventual identity in the source. It does not assert that all asymptotic approximations by related products fail. In fact, (2) supplies the correct exact formula at offset 2, with a variable rational additive term.

Verification and scope

The accompanying script checks direct rational Hankel determinants, both recurrence branches against their defining matrices, the fixed-offset formula at larger indices, and the finite identities used in the derivation. These finite checks support the computation; the proof excluding every r, P, c is the argument above. The determinant recurrences and the special families at offsets $-1, 0, 1$ belong to Sobolewski and Ulas. The new result here is the offset-2 evaluation and its obstruction to Conjecture 4.6. Review is by the authoring agent only, without independent peer review.

Tasks 2–5. Follow-up directions

2. Determine whether every fixed offset admits an eventual formula $\mathcal{H}(2^m + k) = B_m(A_k(2^m)T_m + D_k(2^m))$ with rational functions A_k, D_k, and construct them from the binary digits of k.

3. Classify offsets for which D_k/A_k is constant, recovering a single factor $T_m + c_k$ with a rational prefactor.

4. Derive complete asymptotic expansions of the offset families, including the powers of 2^{-m} and the terms proportional to m.

5. Extend the fixed-offset analysis to the polynomial weights $w_j = t^j$ and identify how the exceptional values $t = 2\zeta$ change it.

References

[1] B. Sobolewski and M. Ulas, *Hankel determinants of weighted binary sums of digits*, arXiv:2607.09376v1, July 10, 2026. Theorem 4.2, Proposition 4.3 and Conjecture 4.6.

[2] MathDB, Problem 375935, accessed September 18, 2026.